CUSTOMER RETENTION & CHURN ANALYSIS

Future Interns — Data Science & Analytics Internship

Task 2: Customer Retention & Churn Analysis

1. Project Overview

Customer retention is an important factor in business growth and profitability. Understanding why customers leave helps businesses identify customer groups that may require additional support and develop strategies to improve customer satisfaction and loyalty.

This project analyzes customer churn using the Telco Customer Churn dataset. The objective is to understand customer behavior, measure churn and retention rates, identify patterns across customer segments, and present the findings through an interactive Power BI dashboard.

2. Project Objectives

The main objectives of this project are:

- Measure the overall customer churn and retention rates.
- Analyze churn across different contract types and payment methods.
- Investigate the relationship between customer tenure and churn.
- Compare churn rates across internet service and customer support categories.
- Examine monthly charges for customers who left and customers who stayed.
- Build an interactive dashboard to support business decision-making.
- Develop data-driven recommendations to improve customer retention.

3. Dataset Description

The project uses the Telco Customer Churn dataset, named WA_Fn-UseC_-Telco-Customer-Churn.csv.

The dataset contains 7,043 customer records and 21 columns. It includes information about customer demographics, account details, subscribed services, billing, contract types, tenure, monthly charges, total charges, and churn status.

The key columns used in the analysis include:

- customerID: Unique customer identifier.
- gender, SeniorCitizen, and Dependents: Customer characteristics.
- tenure: Number of months the customer has stayed.

- Contract: Customer contract type.
- PaymentMethod: Customer payment method.
- InternetService: Type of internet service.
- TechSupport and OnlineSecurity: Subscribed support and security services.
- MonthlyCharges: Monthly amount charged to the customer.
- TotalCharges: Total charges accumulated by the customer.
- Churn: Indicates whether the customer left the service.

The dataset is suitable for examining churn patterns and comparing customer segments.

4. Tools and Technologies

The following tools were used in the project:

- **Microsoft Excel:** Dataset inspection, cleaning review, validation, and tabular reporting.
- **Python:** Data cleaning, data preparation, KPI calculations, and customer segment analysis using pandas and openpyxl.
- **Power BI:** Interactive visualizations, KPI cards, filters, and dashboard development.
- **Git and GitHub:** Project version control, documentation, and portfolio presentation.

5. Data Cleaning and Validation

The original dataset was inspected and prepared before analysis. The cleaned dataset retains all 7,043 customer records and all 21 original columns.

The main data preparation steps included:

1. Removing unnecessary leading and trailing whitespace from text fields.
2. Converting the TotalCharges column into a numeric data type.
3. Identifying 11 blank values in TotalCharges.
4. Filling those 11 blank values with zero because the affected customers had a tenure of zero months.
5. Checking data types and validating important fields.
6. Preserving customer identifiers and churn labels for accurate analysis.
7. Maintaining a data quality report and cleaning summary for documentation.

The cleaned dataset was saved separately from the original raw dataset to preserve the source data.

6. Key Performance Indicators

The analysis focuses on the following KPIs:

Total Customers: The total number of unique customers in the dataset.

Churned Customers: The number of customers whose churn status is Yes.

Retained Customers: The number of customers whose churn status is No.

Churn Rate: The percentage of customers who left the service.

Formula:

Churn Rate = (Churned Customers / Total Customers) × 100

Retention Rate: The percentage of customers who remained with the service.

Formula:

Retention Rate = (Retained Customers / Total Customers) × 100

Average Tenure: The average number of months customers stayed with the service.

Overall Dataset Results

KPI	Result
Total Customers	7,043
Churned Customers	1,869
Retained Customers	5,174
Churn Rate	26.54%
Retention Rate	73.46%

The results indicate that approximately one in four customers in the dataset left the service. The remaining customers represent the larger retained customer group.

7. Customer Churn Analysis

7.1 Churn by Contract Type

The analysis compares churn rates across month-to-month, one-year, and two-year contracts.

Contract Type	Churn Rate
Month-to-month	42.71%
One year	11.27%
Two year	2.83%

Month-to-month customers have the highest observed churn rate, while customers on two-year contracts have the lowest.

This pattern suggests that customers on shorter commitments may require additional retention attention. However, the observed relationship does not establish that contract duration alone causes churn.

7.2 Churn by Customer Tenure

Customer tenure is grouped into four ranges:

- 0–12 months
- 13–24 months
- 25–48 months
- 49–72 months

The purpose of this analysis is to compare churn rates among customers at different stages of their relationship with the company.

The resulting Power BI chart can help identify tenure groups with higher churn rates and inform retention initiatives. The final segment-specific rates should be taken directly from the completed analysis and dashboard.

7.3 Churn by Payment Method

The analysis compares churn across electronic check, mailed check, bank transfer (automatic), and credit card (automatic).

The observed churn rate for customers using electronic checks is approximately 45.29%.

This identifies electronic-check customers as a segment worth investigating. Further analysis would be needed to understand whether payment preferences are associated with other factors, such as contract type, tenure, or service experience.

7.4 Churn by Internet Service

Customers are categorized by internet service type: DSL, fiber optic, or no internet service.

Comparing churn rates across these groups helps identify differences in customer retention between service categories. These results can guide further investigation into service quality, customer expectations, and pricing.

7.5 Churn by Tech Support and Online Security

The analysis examines churn rates for customers with and without tech support and online security, while retaining the no-internet-service category where applicable.

These comparisons help investigate whether support and security subscriptions are associated with different churn patterns. The findings should be interpreted as associations rather than proof that these services directly prevent churn.

7.6 Monthly Charges by Churn Status

Average monthly charges are compared between customers who left and those who stayed.

This comparison helps assess whether the average monthly billing amount differs between the two groups. The final average values should be taken from the Power BI visual or calculated analysis output rather than entered manually.

8. Power BI Dashboard

The Power BI report contains two pages.

Page 1: Customer Overview

The first page presents the overall customer picture through:

- Total Customers KPI card.
- Churn Rate KPI card.
- Retention Rate KPI card.
- Average Tenure KPI card.
- Churned versus Retained Customers chart.
- Churn Rate by Contract Type chart.
- Customer Distribution by Tenure Group chart.
- Contract, Internet Service, and Payment Method slicers.

This page allows users to review the overall customer base and explore the data interactively.

Page 2: Churn Drivers & Retention

The second page examines churn across customer segments through:

- Churn Rate by Tenure Group.
- Churn Rate by Payment Method.
- Churn Rate by Internet Service.
- Churn Rate by Tech Support.
- Churn Rate by Online Security.
- Average Monthly Charges by Churn Status.

These visuals support comparisons between customer groups and help identify segments that may warrant further investigation.

Dashboard screenshots: Insert the exported screenshots of both Power BI pages here.

9. Business Recommendations

Based on the observed patterns, the following actions are worth considering:

1. **Focus on month-to-month customers:** Investigate the reasons for their higher churn rate and evaluate suitable loyalty or longer-term contract offers.
2. **Review the early customer experience:** Examine onboarding, initial service issues, and customer support during the first year.
3. **Investigate electronic-check customers:** Compare this group with other payment segments to understand whether contract type, tenure, billing concerns, or other factors explain its higher churn rate.
4. **Evaluate support and security services:** Investigate whether customers without these services experience different retention outcomes and whether additional support options could improve satisfaction.
5. **Review pricing and billing:** Compare monthly charges across churn groups and investigate whether price sensitivity or perceived value contributes to customer departures.
6. **Monitor churn KPIs regularly:** Track churn rate, retention rate, and segment-level changes over time.
7. **Use targeted retention strategies:** Design interventions based on customer segments and evaluate their results rather than applying the same approach to every customer.

These recommendations are proposed actions based on the observed patterns. Their effectiveness should be measured through subsequent analysis or controlled business experiments.

10. Limitations

The analysis has several limitations:

- The dataset represents a particular customer sample and may not reflect every telecom business.
- The available data supports comparisons and associations but does not establish causal relationships.
- The dataset does not provide the signup dates required to calculate genuine signup-month customer cohorts.
- Some potential explanations for churn, such as detailed complaints, customer satisfaction scores, and specific reasons for cancellation, are not available in the dataset.
- Recommendations should be validated against additional business information before implementation.

11. Conclusion

This project demonstrates an end-to-end customer churn analysis workflow, from data preparation and validation to exploratory analysis and interactive dashboard development.

The dataset contains 7,043 customers, with 1,869 churned customers and an overall churn rate of 26.54%. The observed results show substantial differences in churn rates across contract types, particularly between month-to-month and longer-term contracts. Electronic-check customers also represent a segment that warrants further investigation.

By combining Excel, Python, and Power BI, the project transforms customer records into structured analysis and interactive visualizations. The resulting insights can help businesses investigate retention risks, prioritize customer segments, and develop evidence-based customer retention strategies.

12. Project Deliverables

The project deliverables include:

- Cleaned customer churn dataset.
- Data quality report and cleaning summary.
- Python analysis script.
- Excel analysis results workbook.
- Power BI dashboard with two report pages.
- Dashboard screenshots.
- Final project report and GitHub documentation.

Project: Future Interns — Data Science & Analytics Internship, Task 2
Title: Customer Retention & Churn Analysis
Tools: Excel, Python, Power BI, Git, and GitHub